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Source: *Climate Change Economics*, 2020, Vol. 11, No. 3, Special Issue: The Advances of Integrated Assessment Modeling (IAM) for Climate Change in China (2020), pp. 1-29

Published by: World Scientific Publishing Co., Inc.

Stable URL: <https://www.jstor.org/stable/10.2307/27130668>

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CAN CARBON TAX COMPLEMENT EMISSION TRADING SCHEME? THE IMPACT OF CARBON TAX ON ECONOMY, ENERGY AND ENVIRONMENT IN CHINA

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Accepted 29 April 2020

Published 16 July 2020

The problems of excessive CO₂ emissions and global warming caused by human activities are becoming more serious. Carbon Tax (CT) and Emission Trading Scheme (ETS) are popular emission mitigation mechanisms. This paper establishes four counter-factual (CF) scenarios with different CT rate, and constructs a dynamic recursive computable general equilibrium (CGE) model, named China Energy-Environment-Economy Analysis (CEEEA) model, to study the impact of different CT rate on the economy, energy and environment. The results indicate that if CT complement ETS, and the cap of ETS is based on grandfathering method, the carbon trading price will reduce due to the changes in carbon allowances demand and supply. CT can share the mitigation pressure from ETS coverages into non-ETS coverages. When CT complement ETS but nothing is changed in mechanism of emission trading, the total emission mitigation effect will reduce slightly but the mitigation cost will reduce significantly. All in all, using CT as the supplement is a good mitigation strategy to release Gross Domestic Product (GDP) loss. But if we want to get more mitigation effect, rising CT rate or a stricter carbon cap may help.

Keywords: Computable general equilibrium (CGE) model; emission trading scheme (ETS); carbon dioxide tax (CT); CO₂ emission.

1. Introduction

Energy shortage, environmental degradation and global warming are becoming global issues since the industrial revolution. Greenhouse Gas (GHG) emission is the cause. There is a strong correlation between carbon dioxide (CO₂) emission and other GHG emissions. Many scholars have made several efforts to address issues like global warming (Lin and Jia, 2018a; Li *et al.*, 2017b). They have worked on methods for sustainable development (Ang and Goh, 2016; Sansaniwal *et al.*, 2017)).

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The researches mainly focus on two aspects: low-carbon economy and low carbon technology. For the former, carbon tax (CT) (Zhang *et al.*, 2016; Mardones and Baeza, 2018), forest carbon sinks (Zhang *et al.*, 2013), green electricity certificate trading system (Bunn and Yusupov, 2015; Suo *et al.*, 2017) and clean development mechanism (Koo, 2017; Chen *et al.*, 2017) (especially Emission Trading Scheme (ETS) (Liu *et al.*, 2016b; Creti and Joets, 2017; Viskovic *et al.*, 2017, Liu *et al.*, 2016c)) are the current feasible policies to reduce emissions. For the latter, smart grid, grid energy storage (Zetterberg, 2014; Kapsali and Anagnostopoulos, 2017), electric vehicle (Requia *et al.*, 2017; Masiero *et al.*, 2017), new power generation technologies (Shrivastava and Prabu, 2016) and carbon capture and storage (Stephens and Justo, 2010; Viskovic *et al.*, 2014) are now relatively key technologies for emissions mitigation. As we all know, both ETS and CT can reduce carbon emission by market or price (Bruvoll and Larsen, 2004; Qian, 2017). This paper will focus on the policy method to reduce CO₂; CT and emission trading.

After years of ETS pilot in Beijing, Chongqing, Guangdong, etc., China finally started a national ETS market in 2017 with only power generation industry involved. The goal of China's ETS is to integrate with the international ETS market, such as EU-ETS (Cui *et al.*, 2016; Venmans, 2012), and New South Wales GHG Abatement Scheme (NSW GGAS) (Passey *et al.*, 2008). Therefore, there will be a strong practical significance in studying ETS in China. As an effective method of CO₂ mitigation, ETS has been widely studied from three aspects: the impact of ETS, the design of ETS and the evaluation of ETS. Previous studies on ETS are shown in Table 1.

Many scholars studied CT from various aspects: design or impact of CT. Renner (2018) examine the welfare effects of different CT rates on income distribution. Ma *et al.* (2018) focused on how to make the optimal decision on order quantity and how to select appropriate suppliers for a manufacturer. Farrell (2017) tried to answer the question of what factors drive inequalities in CT incidence. Hariga *et al.* (2017) developed integrated economic and environmental models for a multi-stage cold supply chain under CT regulation. Yang *et al.* (2017) studied the role of revenue sharing and first-mover advantage in emission abatement with CT and consumer environmental awareness. Dong *et al.* (2017) aimed at forecasting the possible impact of CT on both carbon mitigation and economic loss of 30 Chinese provinces. Insley (2017) developed a model of a profit-maximizing firm with the option to exploit a nonrenewable resource, choosing the timing and pace of development. Ghaith and Epplin (2017) studied the effects of a CT on household electricity use and cost, carbon emissions, and economics of household solar and wind.

As for the mixed policy of CT and ETS, or the analyses of differences and links between CT and ETS, related researches exist from many scholars. The earliest literature dates back to 2001: Ekins and Barker (2001) surveys the literature on, and examples of the current implementation of, carbon taxes and carbon emission permits, and argued that early evaluations of the environmental control effectiveness of carbon taxes have been generally positive. While Ermolieva *et al.* (2010) showed that

Table 1. Previous studies on ETS.

Research direction	Author	Research contents	Methodology
Impact of ETS	Wu <i>et al.</i> (2016)	Economic impacts of ETS	CGE model
	Liu and Fan (2018)	Business perspective to the national greenhouse gases emissions trading scheme	A survey
	Li <i>et al.</i> (2017a)	Impacts of ETS on China's coal-to-materials industry in 2020	Cournot model
	Li and Jia (2016)	The impact of ETS and the ratio of free quota	CGE model
Evaluation of ETS	Jaskolski (2016)	Impact of EU-ETS for CO ₂ combined with sulfur dioxide and nitrogen oxides emission trading mechanism on power technology choice	Market allocation model
	Pearse (2016)	Can emissions trading assist with the task of placing a limit on coal production and consumption?	Analyzing on climate policies and related legislation
	Ren and Lo (2017)	Functioning of ETS and the financial performance of the Chinese carbon market	GARCH model
	Zhu <i>et al.</i> (2017)	Can an ETS promote the withdrawal of outdated capacity in energy-intensive sectors?	Input-output analysis
	Ortas and Alvarez (2015)	The efficacy of the EU-ETS	Wavelet coherence analysis
	Chang and Chang (2016)	Proposing the allocation of CO ₂ emissions increment quotas and carbon intensity mitigation burdens	Information entropy method
	Wang <i>et al.</i> (2017)	Establishing a feasible mechanism to help improve the carbon market for renewable energy application in buildings	Volume verification method
Design of ETS	Lin and Jia (2017)	Explore the impacts of ETS and the most suitable coverage industry for China	CGE model
	Liu <i>et al.</i> (2016a)	Designing an approach to optimize the weights of rules	Particle swarm optimization and genetic algorithm
	Tang <i>et al.</i> (2016)	Penalty policies and subsidy policies in ETS	CGE model

the explicit introduction of uncertainties makes it almost impossible for existing market-based ETS and CT to be environmentally safe and cost-effective. Niu (2015) compared ETS and CT by the experience in different countries and considered that the joint application of them can allocate social resource more effectively. Based on the economic order quantity model, He *et al.* (2015) examined the production lot-sizing issues of a firm under these two regulations, respectively. Chiu *et al.* (2015) found that the economic effects of CT and ETS on energy prices depend on the market structures: when the market structure is characterized by perfect competition, the prices will be equal to each other; when it is by imperfect competition, energy price in CT will be lower. Snyder (2015) designed a new emission regulations system, which game includes the government.

A deputy director of the Climate Department of the National Development and Reform Commission (NDRC) in China disclosed that the NDRC is launching a study on the introduction of a CT in 2020.¹ And in 2020, CT may be introduced in the industries which do not participate in ETS.² However, little research focuses on whether CT can complement ETS, and the impact of different CT rate on energy, environment and the economy. This paper will fill the knowledge gap in this field. The paper establishes a dynamic recursive Computable General Equilibrium (CGE) model and constructs a Business as Usual (BAU) scenario and four counter-factual (CF) scenarios to analyze the impact of different CT rate on energy, environment and the economy. The innovations of this paper are as follow:

- (1) This paper establishes a CGE model to analyze impact of different CT rates in the context of ETS, which provides a reference for CGE modelers on how to set up a CT and carbon emission trading block in CGE model.
- (2) This paper tries to determine if CT can complement ETS as an effective mitigation method and suggests the best option of the CT rate.

The basic structure of this paper is as follows: the introduction and literature review are presented in Sec. 1. CGE model, Social Accounting Matrix (SAM) and model dynamics are introduced in Sec. 2. The scenario design is described in Sec. 3. The simulation results and discussions are provided in Sec. 4. The conclusions and suggestions are proposed in Sec. 5. In order to make this paper more concise and easier to read, the main abbreviations in this paper are shown in Table 2.

2. Methodology

2.1. CEEEA/CGE model

CGE model is widely used in policy analysis (Lin and Ouyang, 2014; Paroussos *et al.*, 2015; Lu *et al.*, 2017). The construction of all CGE models is based on traditional

¹Development and Reform Commission Climate Division: It is studying the introduction of CT after 2020. Available at: <http://money.163.com/16/0926/20/C1TQCVMP00253B0H.html>. accessed on March 2018.

²After 2020, China may impose CT. Available at: http://www.sohu.com/a/109823382_161794. accessed on March 2018.

Table 2. The main abbreviations in this paper.

Abbreviation	Full name	Abbreviation	Full name
ETS	Emissions Trading Scheme	CT	CT
VA	Value-added	SAM	Social accounting matrix
VAE	Value-added and Energy	CGE	Computable general equilibrium
CES	Constant Elasticity of Substitution	BAU	Business as usual scenario
CET	Constant Elasticity of Transformation	CF	Counter-factual scenario
CO ₂	Carbon dioxide	NCT	Scenarios of noncarbon tax
GDP	Gross Domestic Product	LCT	Scenarios of low CT rate
CIOT	China Input–Output Table	MCT	Scenarios of medium CT rate
AEI	Autonomous Energy Efficiency Improvement	HCT	Scenarios of high CT rate

Walras paradigm, which means that the model can be described as a system of simultaneous equations deduced by all actors’ maximizing behavior (Hosoe, 2014). CGE model simulates the behavior of social subject, such as residents, enterprises, government, and foreigners (Bohringer *et al.*, 2017; He and Lin, 2017). In this paper, we construct a system of CGE model, named China Energy-Environment-Economy Analysis (CEEEA) model. CEEEA model consists of five blocks: production block, income-expenditure block, trade block, energy policy block, and macroscopic-closure and market-clearing block. The general framework of the CEEEA/CGE model is illustrated in Fig. 1.

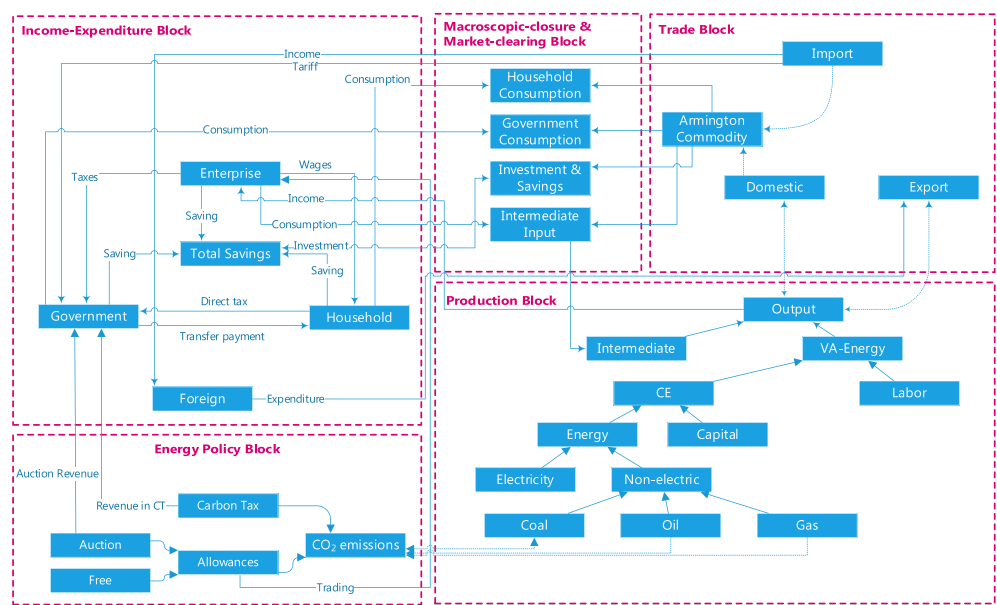


Figure 1. General framework of the CEEEA/CGE model.

2.1.1. *Production block*

We assume that one sector produces only one product in the CGE model. Moreover, in this block, the output consists of policy cost, Value-Added and Energy (VAE) and intermediate input following a Leontief function. VAE is a bundle that consists of CE (capital and energy bundle) and labor following a Constant Elasticity of Substitution (CES) function. The next level is CE bundle, which consists of capital and energy following a CES function. The energy input is constituted by electricity input and nonelectric bundle by a CES function. The nonelectricity energy bundle consists of coal and nonsolid fuel (oil and gas) following a CES function. Others are the same as these CES functions above. Note that intermediate input here is all the intermediate input except energy intermediate, as this paper puts the integration of energy input into CE bundle for studying this kind of issues.

2.1.2. *Income-expenditure block*

The four social subjects are government, enterprise, household (both rural residents and urban residents), and foreigner. The CGE model embodies the balance and relationship between the four subjects. For government, it gets fiscal revenue through direct tax, indirect tax, and tariff; and all the revenue are used for transfer payments, consumption and savings. For domestic enterprises, they get sales revenue from consumption by government, residents, other enterprises and foreigners to support their own expenditure: indirect tax, households' income, and savings. For the households, they get income through remunerations from enterprises and transfer payments by government, and the income is equal to the sum of their consumption, direct tax, and savings.

2.1.3. *Trade block*

CGE model assumes that both domestic and foreign products of an industry are homogeneous. Thus, for one kind of product, imports, and exports cannot exist at the same time. However, imports and exports do exist at the same time in one type of commodity in the real world. Therefore, like most studies in the literature, the Armington assumption is introduced into the CGE model (Lin and Li, 2012; Li *et al.*, 2018) by using CES and Constant Elasticity of Transformation (CET) functions to simulate import and export in the real world.

2.1.4. *Energy-policy block*

This block first introduces the method of integrating energy and emission data into CGE model. And then introduces how to integrate ETS and CT into CGE model. In the production block, this paper eliminates energy input from intermediate input and puts these inputs into CES production function, as Leontief production function is not suitable for such study relative to energy and environmental economics. Intermediate input (energy input) is related to energy consumption. According to AIM/CGE

(Fujimori *et al.*, 2012), the value of energy input, the physical amount of energy input and carbon dioxide emissions are linked by three coefficients shown in Eqs. (1)–(4).

CGE model assumes that ETS market is a perfect competitive market; hence, the auction price of ETS is equal to the trading price of carbon emissions allowance in equilibrium state in modeling. The government will transfer all revenue from CT and ETS to the residents by tax reduction. The rate of Free Allowance (FA) will be 0.9 in the period 2018–2030, and this is set in accordance with the seven pilot cities in China as well as the period I in EU–ETS. In addition, this model assumes that the calculation of cap is based on grandfathering method and there are no extra emissions in coverage industries. This block can be explained by the following six equations:

$$EM_i = \gamma^{\text{coal}} \text{ENE_COAL}_i + \gamma^{\text{oil}} \text{ENE_OIL}_i + \gamma^{\text{gas}} \text{ENE_GAS}_i, \quad (1)$$

$$\text{ENE_COAL}_i = \gamma^{\text{coal}} \text{COAL}_i, \quad (2)$$

$$\text{ENE_OIL}_i = \gamma^{\text{oil}} \text{OIL}_i, \quad (3)$$

$$\text{ENE_GAS}_i = \gamma^{\text{gas}} \text{GAS}_i, \quad (4)$$

$$\text{CAP} = \sum_{ei} EM_{ei} = (1 - \omega) \sum_{ei} Z_{ei} \sum_{ei} EM_{ei,t-1} \sum_{ei} Z_{ei,t-1}, \quad (5)$$

$$\text{PLC}_{nei} = p^{\text{tax}} EM_{nei} R_{nei}, \quad (6)$$

where EM_i represents total emission of a sector. COAL_i and NOS_i represents fossil energy input of the sector i . ENE_COAL_i and ENE_NOS_i are the physical amount of energy input. χ^{coal} and χ^{nos} are the coefficients between value and the physical amount of energy input. γ^{coal} and γ^{nos} represent the carbon emission coefficients of coal, oil, and gas. The subscript of ei represents the coverage industries under the ETS, while the subscript of nei represents the industries which are not covered in the ETS. PLC_{ei} and PLC_{nei} depict the policy cost (ETS or CT cost) in sector ei or nei . p^t is carbon price. CA_{ei} denotes CA of the sector ei . FA_{ei} represents the free CO_2 emission allowances in ETS, while far shows the rate of FAs. p^{tax} is CT rate. EM_{nei} denotes CO_2 emissions in sector nei . R_{nei} is an exogenous variable. If the government imposes a CT, R_{nei} is 1, otherwise it is 0.

The method of combination of CT and ETS into CEEEA model is based on the different subscript, ei and nei , in Eqs. (3)–(6), as well as the exogenous dummy variable R_{nei} . This kind of approach can analyze both issue on CT and ETS. Referring to Appendix A for detailed information about R_{nei} for CT and R_{ei} for ETS.

This paper assumes that the allocation of carbon emission allowances is based on grandfathering method and calculated using the output of sector ei in the current period, last year's CO_2 emission intensity and coefficient of decreasing carbon intensity per year. The FA of sector can be expressed by the following equation:

$$\text{FA}_{ei,t} = Z_{ei,t} \times \frac{EM_{ei,t-1}}{Z_{ei,t-1}} \times (1 - \omega) \times \text{far}, \quad (7)$$

where $FA_{ei,t}$ is FA rate of sector ei in current period. $EM_{ei,t-1}$ represents the CO_2 emission of sector ei in last period. ω represents annual decline factor which is set as 1% in accordance with Guangdong's experience (the pilot city in China) ([Carbon Emissions Trading Network, 2016](#)). But in the first three years, ω will be 2% for fast reduction of redundant carbon emissions. $Z_{ei,t}$ and $Z_{ei,t-1}$ represent the sectorial output of sector ei in the current and last period, respectively.

At present, at least 20 countries have imposed carbon taxes. These countries are broadly divided into two categories: the first category, such as Denmark and Netherlands, already have a comprehensive CT system, started the implementation of the CT system earlier than others, and have better policy efforts. The second category consists of countries that levied CT in the context of a joint global emission mitigation, but implementation is not adequate. Except for the rate of CT and industry coverage, other mechanisms of CT are modeled following the systems of the first category countries — Denmark and the Netherlands: the CT rate is fixed and will be paid in the form of energy tax. CT can be expressed by the following equations:

$$PLC_{nei} = p^{\text{tax}} EM_{nei}, \quad (8)$$

where the subscript nei is the industries which will not be covered by ETS. EM_{nei} is the emission by sector nei . PLC_{nei} is the CT policy cost. p^{tax} denotes CT rate.

2.1.5. Macroscopic-closure and market-clearing block

Three principles of market closure are considered in this model: government budget balance, foreign trade balance, and investment-saving balance. The first two balances are introduced in Sec. 2.1.2 As for investment-saving balance, CGE model assumes that all the savings are transformed into investment, which means that total investment is equal to total savings. Two principles are considered in market clearing. One is the market clearing of Armington composite commodity. The other is factor market clearing. The former shows that all Armington commodities are used for consumption of household and government, intermediate input and savings, without surplus. The latter is that there is no unemployment in the market.

2.2. Social accounting matrix

The most important basic data of CGE model is the SAM, which can be compiled by an input-output table. In this paper, the Input-Output Table of China in 2012 (CIOT) is used. The energy data of SAM is from China Statistical Yearbook (NBS, 2014) and the economic data is from China Input-Output Association (CIOA) ([CIOA, 2010](#)). The SAM is balanced by using SG-RAS method ([Wang et al., 2012](#)). An important point worth noting is that CO_2 emissions in this paper is only from fossil fuel combustion, implying this paper does not consider microbial decomposition, animal and plant respiration.

According to China's plan, chemicals, steel, cement, electricity, construction, and transportation sectors may be the first batch of industries to participate in the carbon

Table 3. Description and coverage of sector classification and population classification.

Abbreviation	Industries	Coverage
AGR	Agriculture, forestry, animal husbandry and fishery	—
COL	Coal mining and washing industry	—
OIL	Petroleum exploitation	—
GAS	Natural gas exploitation	—
REF	Refined oil	Yes
CMC	Chemicals	Yes
MTL	Metal smelting and rolling products	Yes
MTP	Metal products	—
ThP	Thermal power generation	Yes
HyP	Hydropower generation	Yes
WdP	Wind power generation	Yes
NcP	Nuclear generation	Yes
SoP	Solar power generation	Yes
TRA	Transportation, warehousing and postal services	Yes
CST	Construction	Yes
RST	Real estate	—
OTH	Other industry	—
SER	Services	—
RUR	Rural residents	—
CTZ	Urban residents	—

market. Thus, we reclassify the 139 sectors in the CIOT, as shown in Table 3. In addition, according to the experiences of the 7 ETS pilot cities, this paper assumes that the following industries will be covered in China’s ETS (Guangdong PDRC, 2015) (see Table 3).

2.3. Model dynamics

Capital depreciation is determined by the capital stock of the current period and investment. Capital stock is endogenous except for the first period, while investment is endogenous. Labor endowment is exogenous and determined by *National Population Development Plan(2016–2030)* (CPGPRC, 2017). Autonomous Energy Efficiency Improvement (AEEI) in CGE model is considered in this study according to [Li et al. \(2017\)](#). Please refer to Appendix B for detailed information.

3. Scenario Design

On 24 September 2016, the Climate Change Division of the NDRC of China stated that after 2020, CT may be levied on enterprises that are excluded in the carbon market system (Tanpaifang.com, 2016). Thus, this paper assumes that carbon emission trading will be established in 2018 and CT will be implemented in the year 2020. If CT policy

Table 4. Scenario design of carbon prices (Unit: yuan/ton).

Scenario	Establishing ETS	The rate of CT
BAU	—	—
NCT	Yes	—
LCT	Yes	10
MCT	Yes	20
HCT	Yes	50

is applied, sectors who are excluded from ETS will be involved in CT policy. And according to some relevant literatures (Li and Jia, 2017), we set CT rate to three level: 10, 20 and 50 yuan for low, medium and high CT rate (2012 price level). In order to better compare and analyze the impact of CT, we also set a scenario without CT from 2018–2030. The scenario design of carbon prices is depicted in Table 4.

4. Results and Discussion

4.1. Economic impact

4.1.1. ETS price

Figure 2 shows us the ETS price during 2018–2030. The CT will increase from 116.50 CNY in 2018 to 339.14 CNY in 2020. Then the increasing rate of ETS price will reduce due to decreasing technological progress (as shown in the dynamic method in Appendix B). It is indicated that rapid economic development and technological progress will increase the ETS price significantly. Back to our study, we find that CT policy will reduce ETS price. Why does this happen? The following results will gradually solve the puzzle. The logic is: the CT will increase the energy price significantly especially for coal products (Sec. 4.1.3). The demand for energy

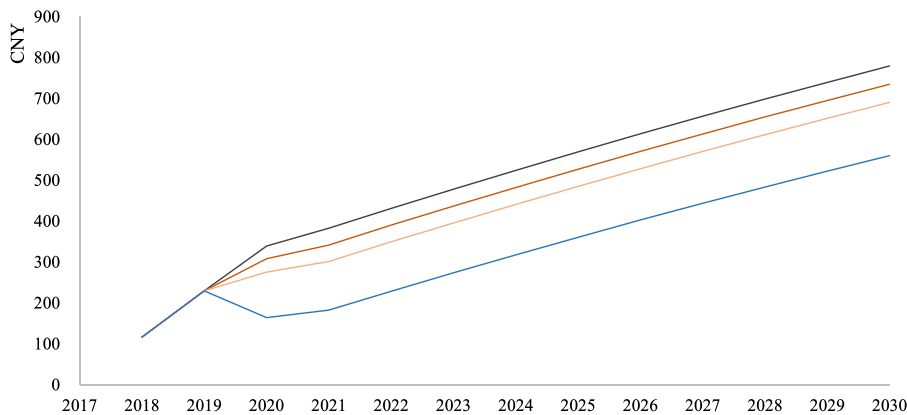


Figure 2. ETS price changes during 2018–2030.

consumption will reduce in all industries due to the high energy cost, as well as the demand for carbon allowances. However, the total cap in ETS market is based on the carbon intensity in last period and output in current period (Sec. 2.1.4), which means no dramatic changes in the allowance supply of ETS market so that ETS price will fall due to CT policy complementing.

4.1.2. GDP

The GDP variation in all CF scenarios compared with the BAU scenario during 2018–2030 is illustrated in Fig. 3. GDP in NCT scenarios will reduce by 0.05–0.19% relative to BAU scenario during 2018–2020. The growth of the negative impact on GDP will be lower during 2021–2030 than 2017–2020, and the negative impact of ETS on GDP will be 0.52% in 2030 NCT scenario. CT policy will be implemented in 2020, and the negative impact of ETS combined with CT on GDP is relatively less than the impact of ETS. GDP in LCT, MCT, and HCT scenarios will decrease by 0.50%, 0.48%, and 0.42% in 2030, respectively. GDP loss will be reduced by 4.11%, 8.12%, and 19.59% in the LCT, MCT, and HCT scenarios compared with NCT scenarios, indicating that CT as the complement of ETS could help to reduce GDP loss. However, alleviating economic pressure does not mean that it is beneficial to emission mitigation, so that we analyze the mitigation effect in Sec. 4.3 and find negative effect of CT due to lower ETS price. And we also analyze output in specific industries in Sec. 4.1.4 to show which part of GDP has increased when CT complement.

4.1.3. Commodity price

The variation in commodity price in the CF scenarios compared with the BAU scenario in 2030 is illustrated in Fig. 4. All commodity prices in CF scenario will increase in 2030 relative to 2030 BAU scenario. ETS will increase commodity prices, especially in ETS coverage industries, such as electricity, refined oil, chemicals, etc. The CT policy

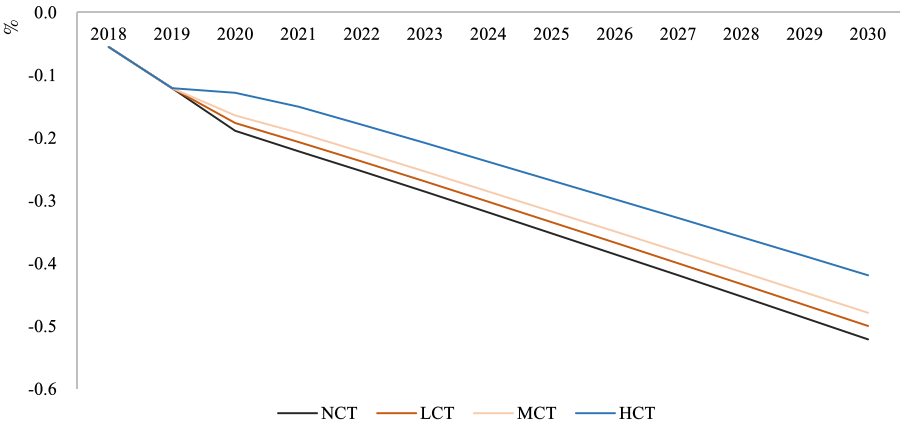


Figure 3. GDP variation of all CF scenarios compared with BAU scenario during 2017–2030.

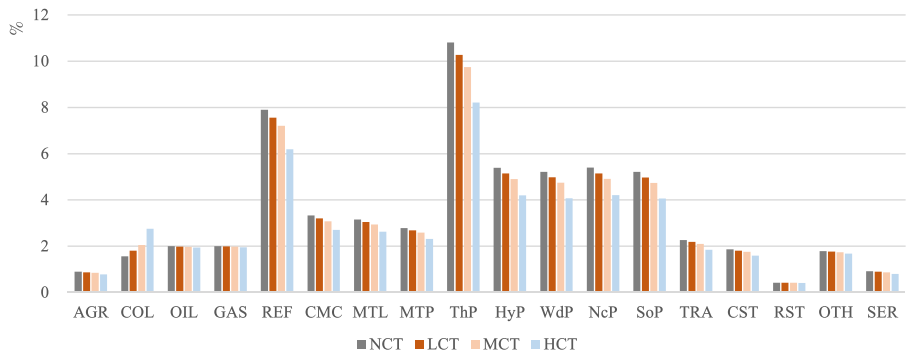


Figure 4. The variation in commodity price in CF scenarios compared with BAU scenario in 2030.

will directly increase the commodity price of energy production (let alone the cost of energy use), especially in coal. The price of coal in HCT scenario will increase by 2.76% compared with the price of BAU scenario, and 1.17% compared with NCT scenario. However, the commodity price will decrease significantly in the ETS coverage industries when CT is implemented as a complement to ETS. For instance, the price of refined oil, chemicals, thermal power generation, and transportation in HCT scenario will only increase by 6.19%, 2.70%, 8.22%, and 1.85% in 2030, respectively compared with the BAU scenario, and will decrease by 1.59%, 0.61%, 2.34%, and 0.42% compared with NCT scenario.

4.1.4. Sectoral output

The variation in sectoral output in all CF scenarios compared with the BAU scenario in 2030 is illustrated in Fig. 5. The output of all industries will decrease in varying degree (except for real estate), by 0.03–14.35%. The output of nonenergy relevant enterprises will decrease by not more than 3%, while the output of energy production and energy conversion industries (such as coal, oil, gas, refined oil, and electricity) will decrease by 7.52–14.35%. We find that higher CT rate is associated with less output reduction,

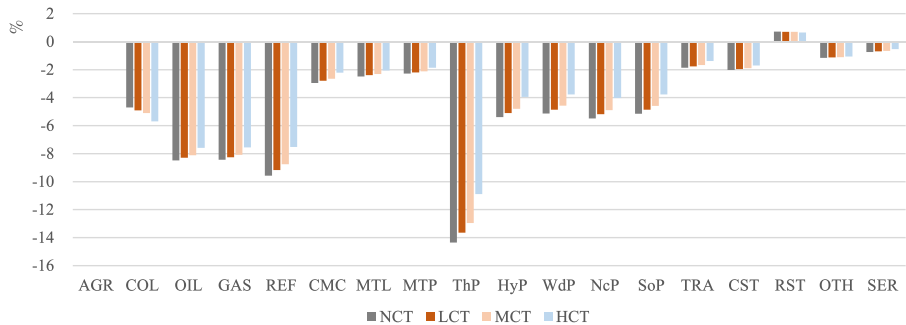


Figure 5. The variation of sectoral output in CF scenarios compared with BAU scenario in 2030.

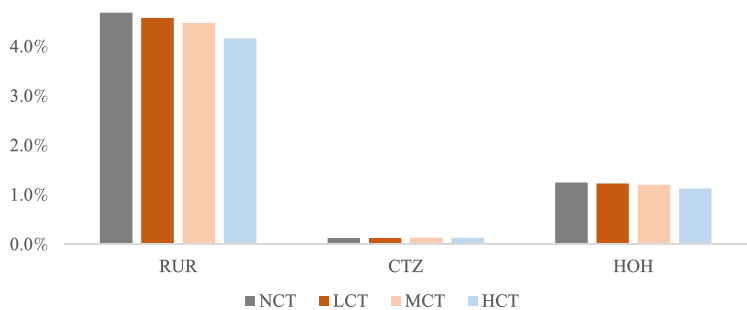


Figure 6. Variation of social welfare in CF scenarios compared with BAU scenario in 2030.

except for coal industries, indicating that complementing CT could significantly reduce the output of coal, the primary fossil energy source in China, but could release the output loss of ETS market. That is the good news for CT, as sectoral output will increase but energy production will decrease.

4.1.5. Social welfare

This paper uses Hicksian equivalent in the form of Laspeyres index to measure social welfare changes (Hosoe *et al.*, 2010), which is shown in Fig. 6. The change of enterprise’s optimal decision caused by ETS or CT leads to the reduction of enterprise’s productivity, so that GDP will reduce. But we find that social welfare will increase when implementing ETS or ETS combining with CT. And the rural residents have a higher increase in welfare. This is based on how the revenues are transferred: we use the approach based on number of population (Lin and Jia, 2018b). But what we are interested in here are the differences between NCT scenarios and the other scenarios. We find that CT will slightly reduce the “double dividend” effect in rural residents but not urban residents. It is because fossil energy consumption accounts for a relatively large proportion of income in rural residents. CT puts the coal price higher than only ETS scenario (Sec. 4.1.3), then leads a lower increase in rural welfare. Although the rural welfare will decrease a little bit, the total social welfare is still increased by 1.25%, 1.23%, 1.21%, and 1.13% in NCT, LCT, MCT, and HCT scenarios, respectively.

4.2. Energy impact

4.2.1. Fossil energy consumption

The variation in fossil energy consumption in the CF scenarios compared with the BAU scenario in 2030 is illustrated in Fig. 7. In NCT scenario, fossil energy consumption of refined oil and thermal power industry will lead the decrease, by 8.93% and 9.57% in 2030, followed by the consumption of energy production industries, such as coal, oil, and gas, by 4.30%, 7.51%, and 7.46%, respectively. Then followed by coverage industries such as transportation and construction. The results are similar to

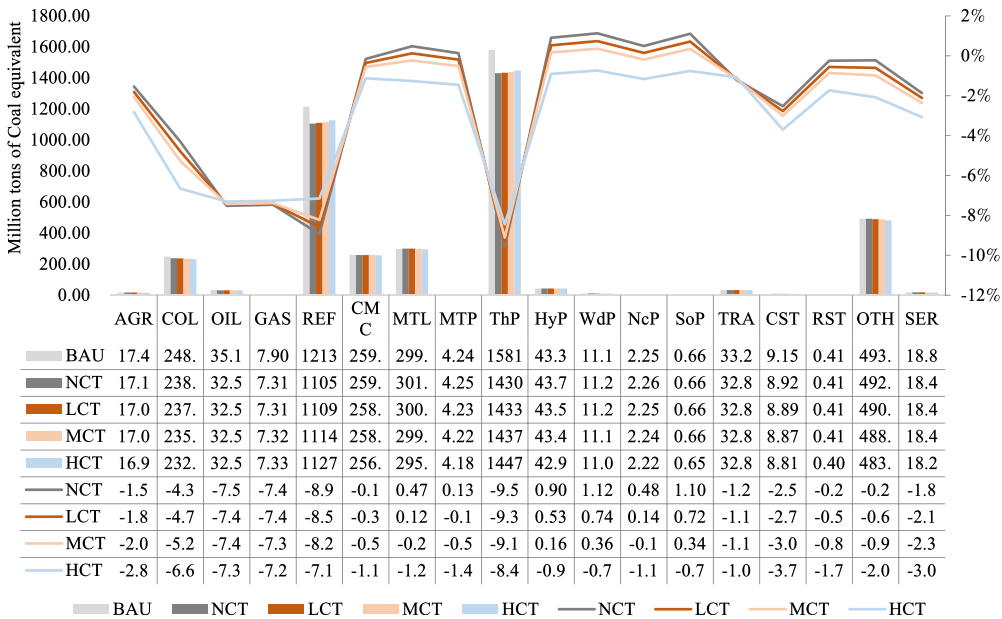


Figure 7. Fossil energy consumption and the variation of fossil energy consumption in CF scenarios compared with BAU scenario in 2030.

the results in domestic output in Sec. 4.1.4, as the production technology cannot be easily changed in the short term. We find that the CT policy could balance the energy consumption among these industries, and in intuitive view, the blue line is straighter than the black line. In a word, the CT will decrease the burden of ETS coverage industries and spread the pressure of emission reduction equally to the society.

4.2.2. Electricity production

The level of and variation in electricity production in the CF scenarios compared with the BAU scenario in 2030 is illustrated in Fig. 8. Due to the resource endowment, China is a primary country using coal as the energy source. This paper finds that ETS and CT can reduce the proportion of thermal power generation to some extent. However, we also find when CT policy combines ETS, the electricity production will increase as well as the proportion of thermal generation compared with NCT scenario, but the impact is limited. In all, CT and ETS can reduce fossil energy consumption and help for cleaner generation.

4.3. Environment impact

4.3.1. CO₂ emission mitigation

The CO₂ emission mitigation in all the scenarios during 2017–2030 is illustrated in Table 5. ETS will reduce CO₂ emission by 5.80 Billion tons of CO₂ (Bt-CO₂) during 2018–2030, or 17.57% of global emission in 2018. The cumulative CO₂ emission

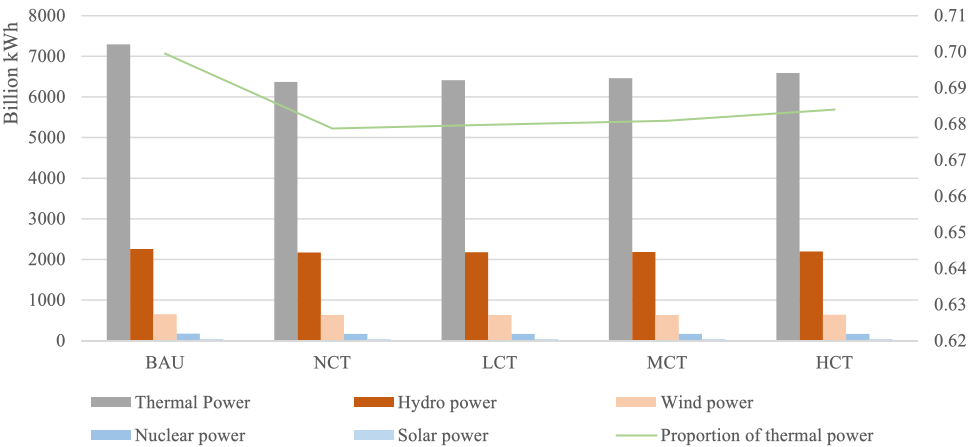


Figure 8. Electricity production and the variation of electricity production in CF scenarios compared with BAU scenario in 2030.

Table 5. Emission mitigation in all CF scenarios during 2018–2030 relative to BAU scenario (Unit: Billion tons of CO₂).

Year	NCT	LCT	MCT	HCT
2018	0.09	0.09	0.09	0.09
2019	0.18	0.18	0.18	0.18
2020	0.28	0.28	0.27	0.25
2021	0.32	0.32	0.31	0.29
2022	0.37	0.36	0.35	0.34
2023	0.41	0.40	0.40	0.38
2024	0.46	0.45	0.44	0.43
2025	0.50	0.49	0.49	0.47
2026	0.55	0.54	0.53	0.51
2027	0.59	0.58	0.58	0.56
2028	0.64	0.63	0.62	0.60
2029	0.68	0.68	0.67	0.65
2030	0.73	0.72	0.71	0.69
Total	5.80	5.73	5.66	5.45

mitigation in the LCT, MCT, and HCT scenarios during 2018–2030 will be 5.73 Bt-CO₂, 5.66 Bt-CO₂, and 5.45 Bt-CO₂, respectively, or 17.36%, 17.15%, and 16.51% of global emission in 2018. We find that emission trading and CT can significantly reduce CO₂ emission. However, the mitigation effect will be slower when implementing CT. The higher the CT rate, the lower CO₂ emission mitigation.

Why will the emission mitigation effect be reduced in the context of CT combining compared with single ETS policy? Then we explore the CO₂ mitigation among different subjects in 2030, as shown in Table 6. ETS coverage industries account for the greatest CO₂ mitigation. While the mitigation effect in ETS coverage industries will

Table 6. CO₂ mitigation among different subjects in 2030 (Unit: Million tons of CO₂).

	ETS coverage	CT coverage	Households	Total
NCT	693.59	41.98	−6.83	728.75
LCT	676.96	50.51	−6.00	721.47
MCT	660.44	58.92	−5.18	714.17
HCT	611.49	83.44	−2.77	692.15

reduce due to complementing of CT, however, the mitigation effect in CT coverage (or non-ETS coverage) sectors will increase accordingly, then leads to a slight reduction of total mitigation effect. The reason is quite simple now. On the one hand, as ETS price going down in the context of CT policy, the mitigation effect of ETS will decrease; on the other hand, CT will change the behavior of enterprises that it covers so that these enterprises will reduce more emission in LCT, MCT and HCT scenarios than that in NCT scenario.

Note that CO₂ emissions we presented here are caused by fossil energy consumption, rather than production. In Sec. 4.1.4, the results indicate that energy production will further decrease in HCT scenario. It doesn't mean that CO₂ emission in HCT scenario is lower than that in NCT scenario because of the international trade. Many researchers argue that CT will stimulate energy import while ETS will not (Strand, 2013). This paper finds similar results but does not present international trade in energy products here due to the limited space.

4.3.2. CO₂ emission intensity

The CO₂ emission intensity (CI) in all the scenarios during 2020–2030 is shown in Fig. 9. CI will be 0.140 ton-CO₂/thousand CNY in 2017 under the BAU scenario and

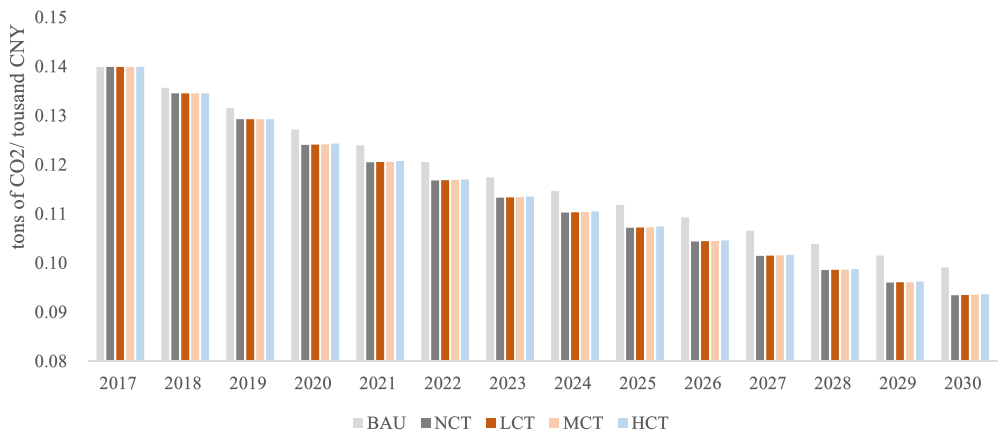


Figure 9. CO₂ emission intensity in all scenarios during 2020–2030.

will be 0.099 ton-CO₂/thousand CNY in 2030. In the NCT, LCT, MCT, and HCT scenarios, CI will be 0.124 ton-CO₂/thousand CNY in 2020; while it will be 0.093, 0.093, 0.094, and 0.094 ton-CO₂/thousand CNY, respectively, in 2030. We find that the supplementary of CT can hardly impact the carbon intensity significantly. Moreover, the higher the CT rate, the higher the carbon intensity, but the impact is minimal. CT and CT rate will lead to insensitive carbon intensity, because higher CT rate will result in less emission mitigation (seen in Table 5) and also less GDP loss (seen in Fig. 3). However, the magnitude of GDP loss is less than the magnitude of emission mitigation, especially in the higher CT rate scenario.

4.3.3. Mitigation cost

The mitigation cost in all CF scenarios during 2018–2030 is shown in Fig. 10. Mitigation cost is the indicator of how much GDP will be reduced by one ton of CO₂ emissions mitigation, which is different from the carbon price (the carbon trading price in ETS market). The mitigation cost will increase 450.08 CNY/tons of CO₂ from 2018 to 844.68 CNY/tons of CO₂ in 2030 in the NCT scenario. In the LCT, MCT and HCT scenario, the average mitigation cost during 2018–2030 will be 645.36, 622.30, 553.62 CNY/tons of CO₂, respectively. We find that the implementation of CT can significantly reduce the mitigation cost to a certain degree. Higher CT rate can help to reduce the mitigation cost. The increasing mitigation cost may be affordable because the mitigation capacity is increasing too. According to Table 5 and Fig. 10, the effects of emission mitigation in the LCT, MCT and HCT scenarios will decrease by 1.18%, 2.37% and 5.98%, respectively compared with the effect in NCT scenario, while the average mitigation cost will decrease by 3.47%, 6.92% and 17.19%, respectively. The mitigation cost is far lower when CT is implemented under the context of ETS. The main reason is that CT reduces the carbon price in ETS market, and puts

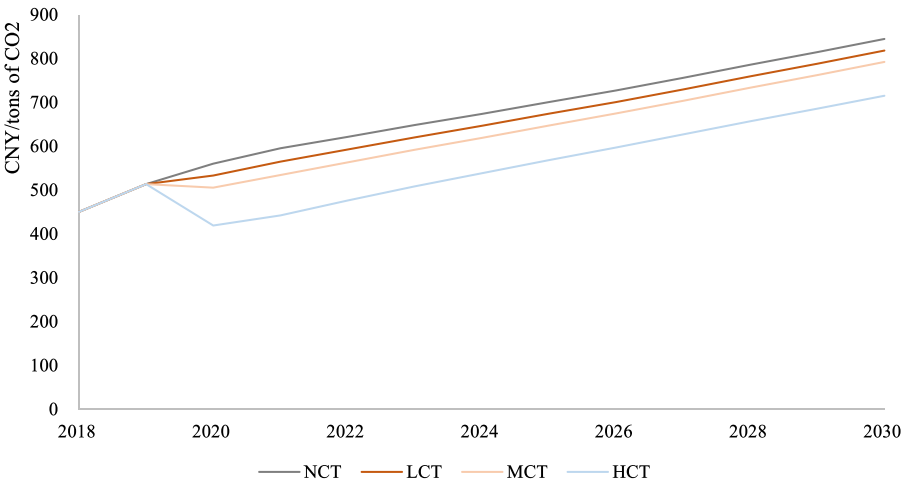


Figure 10. Mitigation cost in all CF scenarios during 2017–2030.

more pressure of emission mitigation on non-ETS coverage industries (Table 6). Decentralized mitigation pressure is conducive to economic development.

5. Conclusions and Policy Implications

This paper establishes four CF scenarios with different CT rate, and constructs a CEEEA/CGE model to study the impact of different CT rate on ETS price, GDP, commodity prices, sectoral output, social welfare, fossil energy consumption, electricity generation, CO₂ emission, the emission intensity and the mitigation cost. We found some objective and critical conclusions. The following conclusions are drawn:

First, not like the common sense, we find that if we introduce CT policy as complement of ETS, the ETS price will fall. Because the CT will lead to the energy price (Fig. 4) increase and stimulate the energy import (Sec. 4.3.1). In energy market, the demand for energy users will decrease due to the increasing price, as well as the demand for carbon allowance. However, the cap of ETS, which is the total supply of carbon allowance, can hardly be changed as the cap is calculated by grandfathering method (Sec. 2.1.4). Thus, the ETS price will go down.

Then we can find complementing CT can help to release GDP loss and sectoral output loss, reduce the pressure of ETS coverage industries, and significantly reduce mitigation cost of the society. However, a good virtue always has its costs: it will slightly increase total emissions, slightly decrease double dividend effect, as well as costs in the energy structure and the carbon intensity. CT can reduce a bit mitigation effect in ETS coverage industries but increase the mitigation effect in non-ETS coverage industries, resulting in a bit less mitigation effect but better mitigation cost (Sec. 4.3.3).

The results show that when CT complements at constant ETS, the total mitigation effect will decrease to some extent. It is easy to tackle this problem: a more rigorous cap. Replacing grandfathering method for calculation cap by the one based on absolute quantities, or a higher reduction requirement per year in grandfathering method can help to increase ETS price when CT complement.

Acknowledgments

This paper is supported by Xiamen University project: Policy and institution of clean, low carbon and sustainable energy system (Grant No. 0610/ZK1108).

Appendix A. Detailed Information of CEEEA Model in this Paper

A.1. Mathematical expression

A.1.1. Production block

$$\text{NOE}_i = \alpha_i^{\text{noe}} [\delta_i^{\text{noe-coal}} \text{COAL}_i^{\rho_i^{\text{noe}}} + \delta_i^{\text{noe-oil}} \text{OIL}_i^{\rho_i^{\text{noe}}} + (1 - \delta_i^{\text{noe-coal}} - \delta_i^{\text{noe-oil}}) \text{GAS}_i^{\rho_i^{\text{noe}}}]^{1/\rho_i^{\text{noe}}}, \quad (\text{A.1})$$

$$\frac{\text{PCOAL}_i}{\text{POIL}_i} = \frac{\delta_i^{\text{noe-coal}}}{\delta_i^{\text{noe-oil}}} \left(\frac{\text{OIL}_i}{\text{COAL}_i} \right)^{(1-\rho_i^{\text{noe}})}, \quad (\text{A.2})$$

$$\frac{\text{PCOAL}_i}{\text{PGAS}_i} = \frac{\delta_i^{\text{noe-coal}}}{1 - \delta_i^{\text{noe-coal}} - \delta_i^{\text{noe-oil}}} \left(\frac{\text{GAS}_i}{\text{COAL}_i} \right)^{(1-\rho_i^{\text{noe}})}, \quad (\text{A.3})$$

$$\text{NOE}_i \text{PNOE}_i = \text{COAL}_i \text{PCOAL}_i + \text{NOS}_i \text{PNOS}_i, \quad (\text{A.4})$$

$$\text{ENE}_i = \alpha_i^{\text{ene}} [\delta_i^{\text{ene}} \text{ELE}_i^{\rho_i^{\text{ene}}} + (1 - \delta_i^{\text{ene}}) \text{NOE}_i^{\rho_i^{\text{ene}}}]^{1/\rho_i^{\text{ene}}}, \quad (\text{A.5})$$

$$\frac{\text{PELE}_i}{\text{PNOE}_i} = \frac{\delta_i^{\text{ene}}}{1 - \delta_i^{\text{ene}}} \left(\frac{\text{NOE}_i}{\text{ELE}_i} \right)^{(1-\rho_i^{\text{ene}})}, \quad (\text{A.6})$$

$$\text{ENE}_i \text{PENE}_i = \text{ELE}_i \text{PELE}_i + \text{NOE}_i \text{PNOE}_i, \quad (\text{A.7})$$

$$\text{CE}_i = \alpha_i^{\text{ce}} [\delta_i^{\text{ce}} \text{ENE}_i^{\rho_i^{\text{ce}}} + (1 - \delta_i^{\text{ce}}) \text{CAP}_i^{\rho_i^{\text{ce}}}]^{1/\rho_i^{\text{ce}}}, \quad (\text{A.8})$$

$$\frac{\text{PENE}_i}{\text{PCAP}_i} = \frac{\delta_i^{\text{ce}}}{1 - \delta_i^{\text{ce}}} \left(\frac{\text{CAP}_i}{\text{ENE}_i} \right)^{(1-\rho_i^{\text{ce}})}, \quad (\text{A.9})$$

$$\text{CE}_i \text{PCE}_i = \text{ENE}_i \text{PENE}_i + \text{CAP}_i \text{PCAP}_i, \quad (\text{A.10})$$

$$\text{VAE}_i = \alpha_i^{\text{vae}} [\delta_i^{\text{vae}} \text{CE}_i^{\rho_i^{\text{vae}}} + (1 - \delta_i^{\text{vae}}) \text{LAB}_i^{\rho_i^{\text{vae}}}]^{1/\rho_i^{\text{vae}}}, \quad (\text{A.11})$$

$$\frac{\text{PCE}_i}{\text{PLAB}_i} = \frac{\delta_i^{\text{vae}}}{1 - \delta_i^{\text{vae}}} \left(\frac{\text{LAB}_i}{\text{CE}_i} \right)^{(1-\rho_i^{\text{vae}})}, \quad (\text{A.12})$$

$$\text{VAE}_i \text{PVAE}_i = \text{CE}_i \text{PCE}_i + \text{LAB}_i \text{PLAB}_i, \quad (\text{A.13})$$

$$\text{INT}_{i,j} = a_{i,j}^{\text{INT}} Z_j, \quad (\text{A.14})$$

$$\text{VAE}_j = a_j^{\text{vae}} Z_j, \quad (\text{A.15})$$

$$\text{PZ}_j = a_j^{\text{vae}} \text{PVAE} + \sum_i a_{i,j}^{\text{INT}} \text{PQ}_i. \quad (\text{A.16})$$

A.1.2. Income-expenditure block

$$\text{SP}_l = ss_l^p \sum_i (\gamma_l^{\text{lab}} \text{LAB}_i \cdot \text{PLAB}_i + \gamma_l^{\text{cap}} \text{CAP}_i \cdot \text{PCAP}_i), \quad (\text{A.17})$$

$$\text{SG} = ss^g \left(\sum_l \text{TD}_l + \sum_i \text{TZ}_i + \sum_i \text{TM}_i \right), \quad (\text{A.18})$$

$$\text{XG}_i = \frac{\mu_i}{\text{PQ}_i} \left(\sum_l \text{TD}_l + \sum_i \text{TZ}_i + \sum_i \text{TM}_i - \text{SG} \right), \quad (\text{A.19})$$

$$\text{XP}_{i,l} = \frac{\beta_{i,l}^{\text{xp}}}{\text{PQ}_i} \left(\sum_i (\gamma_l^{\text{lab}} \text{LAB}_i \cdot \text{PLAB}_i + \gamma_l^{\text{cap}} \text{CAP}_i \cdot \text{PCAP}_i) - \text{SP}_l - \text{TD}_l \right), \quad (\text{A.20})$$

$$\begin{aligned} \text{TD}_l &= \tau_l^d \sum_i (\gamma_l^{\text{lab}} \text{LAB}_i \cdot \text{PLAB}_i + \gamma_l^{\text{cap}} \text{CAP}_i \cdot \text{PCAP}_i) \\ &\quad - \frac{\text{FF}_l^{\text{lab}} + \text{FF}_l^{\text{cap}}}{\sum_l (\text{FF}_l^{\text{lab}} + \text{FF}_l^{\text{cap}})} \times \sum_i \text{PLC}_i, \end{aligned} \quad (\text{A.21})$$

$$\text{TZ}_i = \tau^z \text{PZ}_i Z_i + \text{PLC}_i, \quad (\text{A.22})$$

$$\text{TM}_i = \tau^m \text{PM}_i M_i. \quad (\text{A.23})$$

A.1.3. Trade block

$$\text{PE}_i = \varepsilon \text{PWE}_i, \quad (\text{A.24})$$

$$\text{PM}_i = \varepsilon \text{PWM}_i, \quad (\text{A.25})$$

$$\sum_i \text{PWE}_i E_i + \text{SF} = \sum_i \text{PWM}_i M_i, \quad (\text{A.26})$$

$$Q_i = \gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{1/\eta_i}, \quad (\text{A.27})$$

$$M_i = \left[\frac{\gamma_i^{\eta_i} \delta m_i \text{PQ}_i}{(1 + \tau_i^m) \text{PM}_i} \right]^{\frac{1}{1-\eta_i}} Q_i, \quad (\text{A.28})$$

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i \text{PQ}_i}{\text{PD}_i} \right]^{\frac{1}{1-\eta_i}} Q_i, \quad (\text{A.29})$$

$$Z_i = \theta_i (\xi e_i E_i^{\phi_i} + \xi d_i D_i^{\phi_i})^{\frac{1}{\phi_i}}, \quad (\text{A.30})$$

$$E_i = \left[\frac{\theta_i^{\phi_i} \xi e_i [(1 + \tau_i^z) + \text{PLC}_i / \text{PZ}_i / Z_i] \text{PZ}_i}{\text{PE}_i} \right]^{\frac{1}{1-\phi_i}} Z_i, \quad (\text{A.31})$$

$$D_i = \left[\frac{\theta_i^{\phi_i} \xi d_i [(1 + \tau_i^z) + \text{PLC}_i / \text{PZ}_i / Z_i] \text{PZ}_i}{\text{PD}_i} \right]^{\frac{1}{1-\phi_i}} Z_i. \quad (\text{A.32})$$

A.1.4. *Energy-Policy block*

$$EM_i = \gamma^{\text{coal}} \text{ENE_COAL}_i + \gamma^{\text{oil}} \text{ENE_OIL}_i + \gamma^{\text{gas}} \text{ENE_GAS}_i, \quad (\text{A.33})$$

$$\text{ENE_COAL}_i = \gamma^{\text{coal}} \text{ENE_OIL}_i, \quad (\text{A.34})$$

$$\text{ENE_OIL}_i = \gamma^{\text{oil}} \text{ENE_OIL}_i, \quad (\text{A.35})$$

$$\text{ENE_GAS}_i = \gamma^{\text{gas}} \text{ENE_GAS}_i, \quad (\text{A.36})$$

$$\text{PLC}_{ei} = p^t (EM_{ei} - \text{FA}_{ei}) R_{ei}, \quad (\text{A.37})$$

$$\text{FA}_{ei} R_{ei} = Z_{ei} \frac{EM_{ei,t-1}}{Z_{ei,t-1}} (1 - \omega) \cdot \text{far} \cdot R_{ei}, \quad (\text{A.38})$$

$$R_{ei} \sum_{ei} EM_{ei} = (1 - \omega) R_{ei} \sum_{ei} Z_{ei} \frac{\sum_{ei} EM_{ei,t-1}}{\sum_{ei} Z_{ei,t-1}}, \quad (\text{A.39})$$

$$\text{PLC}_{\text{nei}} = p^{\text{tax}} EM_{\text{nei}} R_{\text{nei}}. \quad (\text{A.40})$$

A.1.5. *Macroscopic-closure and market-clearing block*

$$\text{XV}_i = \frac{\lambda_i}{\text{PQ}_i} \left(\sum_l \text{SP}_l + \text{SG} + \varepsilon \text{SF} \right), \quad (\text{A.41})$$

$$\mathcal{Q}_i = \sum_l \text{XP}_{i,l} + \text{XG}_i + \text{XV}_i + \sum_j \text{INT}_{i,j}, \quad (\text{A.42})$$

$$\sum_i \text{LAB}_i = \sum_l \text{FF}_l^{\text{lab}}, \quad (\text{A.43})$$

$$\sum_i \text{CAP}_i = \sum_l \text{FF}_l^{\text{cap}}. \quad (\text{A.44})$$

A.1.6. *Objective function*

$$\text{TOTUU} = \sum_l \left(\prod_i \text{XP}_{i,l}^{\alpha_{i,l}} \right). \quad (\text{A.45})$$

A.2. Descriptions of key variables

As there are lots of variables in CEEEA model, this paper only introduces several parts of the key variables but ignore the introduction of several parameters, such as α_i^{noe} , δ_i^{noe} and ρ_i^{noe} in lots of CES functions. The key variables are shown in Table A.1.

A.3. Descriptions of equations

Equations (A.1) to (A.4) are a CES function bundle. The first is a CES production function, the second and the third are the first-order conditions of the CES function,

Table A.1. Descriptions of key variables.

Variables	Descriptions
NOE_i	Non-electric input
$COAL_i$	Coal input
OIL_i	Oil input
GAS_i	Gas input
ELE_i	Electricity input
ENE_i	Energy input
CAP_i	Capital input
LAB_i	Labor input
CE_i	Capital and energy input
VAE_i	Value added and energy input
Z_j	Domestic output
$INT_{i,j}$	Intermediate input
$PNOE_i$	Price of non-electric input
$PCOAL_i$	Price of coal input
$POIL_i$	Price of oil input
$PGAS_i$	Price of gas input
$PELE_i$	Price of electricity input
$PENE$	Price of energy input
$PCAP_i$	Price of capital input
$PLAB_i$	Price of labor input
PCE	Price of capital and energy input
$PVAE_i$	Price of value added and energy input
PZ_j	Price of domestic output
SP_l	Savings of residents l
SG	Savings of the government
XP_i	Consumption of the residents l
XG_i	Consumption of the government
TD_i	Direct tax
TZ_i	Indirect tax
TM_i	Tariffs
E_i	Export
D_i	Domestic goods for domestic consumption
M_i	Import
Q_i	Armington commodity
SF	Trade deficit
PE_i	Price of export
PD_i	Price of domestic goods for domestic consumption
PM_i	Price of import
PQ_i	Price of Armington commodity
PWE_i	Price of export in USD
PWM_i	Price of import in USD
EM_i	CO ₂ emissions
ENE_COAL_i	Coal consumption (Unit: million tonnes of coal equivalent)
ENE_OIL_i	Oil consumption (Unit: million tons of coal equivalent)
ENE_GAS_i	Gas consumption (Unit: million tons of coal equivalent)
PLC_{ei}	ETS cost

Table A.1. (Continued)

Variables	Descriptions
p^t	ETS price
far	Free allowance rate
FA_{ei}	Free part of carbon allowances
PLC_{nei}	CT cost
p^{tax}	CT rate
R_{ei}	Dummy variable (1 if ETS policy is implemented, 0 for others)
R_{nei}	Dummy variable (1 if CT policy is implemented, 0 for others)
XV_i	Investment
FF_l^{lab}	The endowment of labor
FF_l^{cap}	The endowment of capital
TOTUU	Total utility

and the last is an equation to balance the value. From equation (A.1) to (A.13), all of them are CES function bundles in production block at different nest level. Equations from (A.14) to (A.16) describe the output bundle by Leontief production function. Equation (A.13) and (A.14) are the relationships between output and intermediate input, and between output and VAE input. Equation (A.16) is simplified value identity equation. Bringing Equation (A.14) and (A.15) into Equation (A.16) can get an identity equation similar to equation (A.4).

Income and expenditure block describes the behaviors of household savings (A.17), government savings (A.18), household consumption (A.19), government consumption (A.19), direct tax (A.21), indirect tax (A.22), and tariff (A.23). Note that the cost of CT and ETS are in the form of tax levied by government. So, there are several differences in the equations of (A.21) and (A.22).

Equation (A.24) and (A.25) describe the relationship between domestic price and international price. As the international trade is not what we focus on, this paper follows the small country assumption. Equation (A.26) expresses trade deficit. Equation (A.27) to (A.29) are Armington assumption, which describe the behavior of consuming import goods and consuming domestic goods. Equation (A.27) is a CES function and the next two equations are the first-order conditions of it. Like the former equation bundle, equations from (A.30) to (A.32) describe the behavior of domestic enterprises to distribute their product to domestic market and international market. Among them, equation (A.30) is a CET function, and equations (A.31) and (A.32) are the first-order condition of equation (A.30).

Equations (A.33) and (A.36) describe the relationship among fossil fuel consumption, CO2 emissions, and factor inputs in production block. Equations from (A.37) to (A.39) describe the market of ETS. Subscript *ei* is the industries covered by ETS. Equation (40) expresses the carbon tax. Subscript *nei* is the industries not covered by ETS.

Equation (A.41) is mathematical expression of macroscopic closure. Equation (A.42), (A.43) and (A.44) are mathematical expressions of market clearing. Equation (A.45) is utility function, which is also the objective function.

Appendix B. Dynamic Method

All of the variables shown in Appendix A omit subscript t to make the model more readable. Thus, the equation system shown here is a system of static CGE model. This paper uses a recursive method to make the model dynamic. We consider three dynamic parts: population growth, capital accumulation and technological progress. We change some of the exogenous variables to meet the real growth of the society. For population growth, we follow *National Population Development Plan (2016–2030)* ([The Central People’s Government of the People’s Republic of China, 2017](#)). For capital accumulation, this paper uses perpetual inventory method to determine capital stock in each year. For Technological progress, this paper assumes the scale coefficient of CES production function in VAE bundle (α_i^{val}) will increase according to *Medium and Long-term Energy Saving Special Planning* (NDRC, 2005).

Table B.1. AEEI, capital depreciation rate and capital stock of each sector in CGE model^a.

Sector’s Abbreviations ^b	AEEI	Depreciation rate	Capital stock
AGR	0.025	0.050	4516.3
COL	0.006	0.062	1549.6
OIL	0.006	0.065	1174.0
GAS	0.006	0.065	263.4
REF	0.006	0.065	1353.3
CMC	0.015	0.055	6363.6
MTL	0.025	0.055	7085.1
MTP	0.02	0.055	1526.7
ThP	0.025	0.048	6423.2
HyP	0.025	0.048	2028.5
WdP	0.025	0.048	499.7
NcP	0.025	0.048	155.6
SoP	0.025	0.048	29.4
TRA	0.033	0.052	9654.0
CST	0.006	0.055	2992.6
RST	0.01	0.052	30068.9
OTH	0.016	0.055	26228.8
SER	0.023	0.045	38443.5

^aAEEI will be in halve after 2020.
^bSector’s abbreviations can be referred to Table 3.

Table B.2. Population growth rate in this paper.

Year	Population growth rate
2012–2015	0.60%
2016–2020	0.60%
2021–2025	0.21%
2026–2030	0.15%

References

Ang, BW and T Goh (2016). Carbon intensity of electricity in ASEAN: Drivers, performance and outlook. *Energy Policy*, 98, 170–179, doi: 10.1016/j.enpol.2016.08.027.

Bohringer, C, X Garcia-Muros, I Cazcarro and I Arto (2017). The efficiency cost of protective measures in climate policy. *Energy Policy*, 104, 446–454, doi: 10.1016/j.enpol.2017.01.007.

Bruvoll, A and BM Larsen (2004). Greenhouse gas emissions in Norway: Do carbon taxes work? *Energy Policy*, 32(4), 493–505, doi: 10.1016/S0301-4215(03)00151-4.

Bunn, D and T Yusupov (2015). The progressive inefficiency of replacing renewable obligation certificates with contracts-for-differences in the UK electricity market. *Energy Policy*, 82, 298–309. doi: 10.1016/j.enpol.2015.01.002.

Carbon Emissions Trading Network (2016). Carbon emissions quota allocation scheme in Guangdong province. Available at <http://www.tanpaifang.com/zhengcefagui/2016/070954515.html>. Accessed on August 2017.

Chang, K and H Chang (2016). Cutting CO2 intensity targets of interprovincial emissions trading in China. *Applied Energy*, 163, 211–221, doi: 10.1016/j.apenergy.2015.10.146.

Chen, Y, W Hu, P Chen and R Ruan (2017). Household biogas CDM project development in rural China. *Renewable and Sustainable Energy Reviews*, 67, 184–191, doi: 10.1016/j.rser.2016.09.052.

China Input-Output Association (CIOA) (2010). Input-output table. Available at <http://www.iochina.org.cn/Download/xgxz.html>. Accessed on August 2017.

Chiu, FP, HI Kuo, CC Chen and CS Hsu (2015). The energy price equivalence of carbon taxes and emissions trading-Theory and evidence. *Applied Energy*, 160, 164–171, doi: 10.1016/j.apenergy.2015.09.022.

Creti, A and M Joets (2017). Multiple bubbles in the European Union emission trading scheme. *Energy Policy*, 107, 119–130, doi: 10.1016/j.enpol.2017.04.018.

Cui, Q, YM Wei and Y Li (2016). Exploring the impacts of the EU ETS emission limits on airline performance via the dynamic environmental DEA approach. *Applied Energy*, 183, 984–994, doi: 10.1016/j.apenergy.2016.09.048.

Dong, HJ, Y Geng, T Fujita, Z Liu, Y Xie, R Wu, M Fujii, T Masui and L Tang (2017). Exploring impact of carbon tax on China’s CO2 reductions and provincial disparities. *Renewable and Sustainable Energy Reviews*, 77, 596–603, doi: 10.1016/j.rser.2017.04.044.

Ekins, P and T Barker (2001). Carbon taxes and carbon emissions trading. *Journal of Economic Surveys*, 15(3), 325–376.

Ermolieva, T, Y Ermoliev, G Fischer, M Jonas, M Makowski and F Wagner (2010). Carbon emission trading and carbon taxes under uncertainties. *Climate Change*, 103(1–2), 277–289, doi: 10.1007/s10584-010-9910-x.

- Farrell, N (2017). What factors drive inequalities in carbon tax incidence? Decomposing socioeconomic inequalities in carbon tax incidence in Ireland. *Ecological Economics*, 142, 31–45, doi: 10.1016/j.ecolecon.2017.04.004.
- Fujimori, S, T Masui and Y Matsuoka (2012). AIM/CGE [basic] manual. Discussion Paper Series, doi:10.3386/w16827.
- Ghaith, AF and FM Epplin (2017). Consequences of a carbon tax on household electricity use and cost, carbon emissions, and economics of household solar and wind. *Energy Economics*, 67, 159–168, doi: 10.1016/j.eneco.2017.08.012.
- Guangdong Provincial Development and Reform Commission. (2015). Carbon emission quota allocation scheme for Guangdong Province in 2015. Available at <http://www.mzdp.gov.cn/show/index/631/14632>. Accessed on August 2017.
- Hariga, M, R As'ad and A Shamayleh (2017). Integrated economic and environmental models for a multi stage cold supply chain under carbon tax regulation. *Journal of Cleaner Production*, 166, 1357–1371, doi: 10.1016/j.jclepro.2017.08.105.
- He, P, W Zhang, XY Xu and YW Bian (2015). Production lot-sizing and carbon emissions under cap-and-trade and carbon tax regulations. *Journal of Cleaner Production*, 103, 241–248, doi: 10.1016/j.jclepro.2014.08.102.
- He, YD and BQ Lin (2017). The impact of natural gas price control in China: A computable general equilibrium approach. *Energy Policy*, 107, 524–531, doi: 10.1016/j.enpol.2017.05.015.
- Hosoe, N, K Gasawa and H Hashimoto (2010). *Textbook of Computable General Equilibrium Modelling: Programming and Simulations*. New York: St Martin's Press, Palgrave Macmillan, pp. xix, 235 2–xix, 235. doi: 10.1017/CBO9781107415324.004.
- Hosoe, N (2014). Estimation errors in input-output tables and prediction errors in computable general equilibrium analysis. *Economic Modelling*, 42, 277–286, doi: 10.1016/j.econmod.2014.07.012.
- Insley, M (2017). Resource extraction with a carbon tax and regime switching prices: Exercising your options. *Energy Economics*, 67, 1–16, doi: 10.1016/j.eneco.2017.07.013.
- Jaskolski, M (2016). Modelling long-term technological transition of Polish power system using MARKAL: Emission trade impact. *Energy Policy*, 97, 365–377, doi: 10.1016/j.enpol.2016.07.017.
- Kapsali, M and JS Anagnostopoulos (2017). Investigating the role of local pumped-hydro energy storage in interconnected island grids with high wind power generation. *Renewable Energy*, 114(B), 614–628, doi: 10.1016/j.renene.2017.07.014.
- Koo, B (2017). Examining the impacts of feed-in-tariff and the clean development mechanism on Korea's renewable energy projects through comparative investment analysis. *Energy Policy*, 104, 144–154, doi: 10.1016/j.enpol.2017.01.017.
- Li, GY, J Yang, DJ Chen and SY Hu (2017a). Impacts of the coming emission trading scheme on China's coal-to-materials industry in 2020. *Applied Energy*, 195, 837–849, doi: 10.1016/j.apenergy.2017.03.115.
- Li, H, YH Zhao, XY Qiao, Y Liu, C Ye, Y Li, S Wang, ZH Zhang, YF Zhang and JF Weng (2017b). Identifying the driving forces of national and regional CO₂ emissions in China: Based on temporal and spatial decomposition analysis models. *Energy Economics*, 68, 522–538, doi: 10.1016/j.eneco.2017.10.024.
- Li, W and ZJ Jia (2017). Carbon tax, emission trading, or the mixed policy: Which is the most effective strategy for climate change mitigation in China? *Mitigation and Adaptation Strategies for Global Change*, 22(6), 973–992, doi: 10.1007/s11027-016-9710-3.

- Li, W and ZJ Jia (2016). The impact of emission trading scheme and the ratio of free quota: A dynamic recursive CGE model in China. *Applied Energy*, 174, 1–14, doi: 10.1016/j.apenergy.2016.04.086.
- Li, W, ZJ Jia and HZ Zhang (2017c). The impact of electric vehicles and CCS in the context of emission trading scheme in China: A CGE-based analysis. *Energy*, 119, 800–816, doi: 10.1016/j.energy.2016.11.059.
- Li, ZL, HC Dai, L Sun, Y Xie, Z Liu, P Wang and H Yabar (2018). Exploring the impacts of regional unbalanced carbon tax on CO2 emissions and industrial competitiveness in Liaoning province of China. *Energy Policy*, 113, 9–19, doi: 10.1016/j.enpol.2017.10.048.
- Lin, BQ and ZJ Jia (2017). The impact of Emission Trading Scheme (ETS) and the choice of coverage industry in ETS: A case study in China. *Applied Energy*, 205, 1512–1527, doi: 10.1016/j.apenergy.2017.08.098.
- Lin, BQ and ZJ Jia (2018a). Impact of quota decline scheme of emission trading in China: A dynamic recursive CGE model. *Energy*, 149, 190–203, doi: 10.1016/j.energy.2018.02.039.
- Lin, B and Z Jia (2018b). Transfer payments in emission trading markets: A perspective of rural and urban residents in China. *Journal of Cleaner Production*, 204, 753–766. doi: 10.1016/j.jclepro.2018.09.079.
- Lin, BQ and AJ Li (2012). Impacts of removing fossil fuel subsidies on China: How large and how to mitigate? *Energy*, 44(1), 741–749, doi: 10.1016/j.energy.2012.05.018.
- Lin, BQ and XL Ouyang (2014). A revisit of fossil-fuel subsidies in China: Challenges and opportunities for energy price reform. *Energy Conversion and Management*, 82, 124–134, doi: 10.1016/j.enconman.2014.03.030.
- Liu, XB and YB Fan (2018). Business perspective to the national greenhouse gases emissions trading scheme: A survey of cement companies in China. *Energy Policy*, 112, 141–151, doi: 10.1016/j.enpol.2017.10.019.
- Liu, XJ, HZ An, LJ Wang and XL Jia (2016a). An integrated approach to optimize moving average rules in the EUA futures market based on particle swarm optimization and genetic algorithms. *Applied Energy*, 185(2), 1778–1787, doi: 10.1016/j.apenergy.2016.01.045.
- Liu, Y, LY Han, ZQ Yin and KY Luo (2016b). A competitive carbon emissions scheme with hybrid fiscal incentives: The evidence from a taxi industry. *Energy Policy*, 102, 414–422, doi: 10.1016/j.enpol.2016.12.038.
- Liu, Y, XJ Tan, Y Yu and SZ Qi (2016c). Assessment of impacts of Hubei Pilot emission trading schemes in China — A CGE-analysis using TermCO2 model. *Applied Energy*, 189, 762–769, doi: 10.1016/j.apenergy.2016.05.085.
- Lu, YY, Y Liu and MF Zhou (2017). Rebound effect of improved energy efficiency for different energy types: A general equilibrium analysis for China. *Energy Economics*, 62, 248–256, doi: 10.1016/j.eneco.2017.01.010.
- Ma, X, P Ji, W Ho and CH Yang (2018). Optimal procurement decision with a carbon tax for the manufacturing industry. *Computers & Operations Research*, 89, 360–368, doi: 10.1016/j.cor.2016.02.017.
- Mardones, C and N Baeza (2018). Economic and environmental effects of a CO2 tax in Latin American countries. *Energy Policy*, 114, 262–273, doi: 10.1016/j.enpol.2017.12.001.
- Masiero, G, MH Ogasavara, AC Jussani and ML Risso (2017). The global value chain of electric vehicles: A review of the Japanese, South Korean and Brazilian cases. *Renewable and Sustainable Energy Reviews*, 80, 290–296, doi: 10.1016/j.rser.2017.05.148.
- National Bureau of Statistics (NBS) (2014). China Statistical Yearbook (2014). Available at <http://www.stats.gov.cn/tjsj/ndsj/>. Accessed on August 2017.

- National Development and Reform Commission (2005). Medium and Long-term Energy Saving Special Planning. Available at http://www.nea.gov.cn/2011-08/18/c_131057667.htm. Accessed on July 2020.
- Niu, HY (2015). The comparison between international trading system of carbon tax and carbon emission. In *2nd International Conference on Creative Education (ICCE 2015)*, Advances in Education Sciences, London, England, Vol. 11, Pt. 2, pp. 440–444.
- Ortas, E and I Alvarez (2015). The efficacy of the European Union Emissions Trading Scheme: Depicting the co-movement of carbon assets and energy commodities through wavelet decomposition. *Journal of Cleaner Production*, 116, 40–49, doi: 10.1016/j.jclepro.2015.12.112.
- Paroussos, L, P Fragkos, P Capros and K Fragkiadakis (2015). Assessment of carbon leakage through the industry channel: The EU perspective. *Technological Forecasting and Social Change*, 90, 204–219, doi: 10.1016/j.techfore.2014.02.011.
- Passey, R, I MacGill and H Outhred (2008). The governance challenge for implementing effective market-based climate policies: A case study of The New South Wales Greenhouse gas reduction scheme. *Energy Policy*, 36(8), 3009–3018, doi: 10.1016/j.enpol.2008.04.010.
- Pearse, R (2016). The coal question that emissions trading has not answered. *Energy Policy*, 99, 319–328, doi: 10.1016/j.enpol.2016.05.053.
- Qian, HQ, LB Wu and WQ Tang (2017). “Lock-in” effect of emission standard and its impact on the choice of market based instruments. *Energy Economics*, 63, 41–50, doi: 10.1016/j.eneco.2017.01.005.
- Ren, C and AY Lo (2017). Emission trading and carbon market performance in Shenzhen, China. *Applied Energy*, 193, 414–425, doi: 10.1016/j.apenergy.2017.02.037.
- Renner, S (2018). Poverty and distributional effects of a carbon tax in Mexico. *Energy Policy*, 112, 98–110, doi: 10.1016/j.enpol.2017.10.011.
- Requia, WJ, MD Adams, A Arain, P Koutrakis and M Ferguson (2017). Carbon dioxide emissions of plug-in hybrid electric vehicles: A life-cycle analysis in eight Canadian cities. *Renewable and Sustainable Energy Reviews*, 78, 1390–1396, doi: 10.1016/j.rser.2017.05.105.
- Sansaniwal, SK, MA Rosen and SK Tyagi (2017). Global challenges in the sustainable development of biomass gasification: An overview. *Renewable and Sustainable Energy Review*, 80, 23–43, doi: 10.1016/j.rser.2017.05.215.
- Shrivastava, A and V Prabu (2016). Thermodynamic analysis of solar energy integrated underground coal gasification in the context of cleaner fossil power generation. *Energy Conversion and Management*, 110, 67–77, doi: 10.1016/j.enconman.2015.12.009.
- Snyder, BF (2015). Tax and trade: A hybrid climate policy instrument to control carbon prices and emissions. *Climate Policy*, 15(6), 743–750, doi: 10.1080/14693062.2014.965655.
- Strand, J (2013). Strategic climate policy with offsets and incomplete abatement: Carbon taxes versus cap-and-trade. *Journal of Environmental Economics and Management*, 66, 202–218, doi: 10.1016/j.jeem.2013.03.002.
- Stephens, JC and S Jiusto (2010). Assessing innovation in emerging energy technologies: Socio-technical dynamics of carbon capture and storage (CCS) and enhanced geothermal systems (EGS) in the USA. *Energy Policy*, 38(4), 2020–2031, doi: 10.1016/j.enpol.2009.12.003.
- Suo, C, YP Li, SW Jin, J Liu, YF Li and RF Feng (2017). Identifying optimal clean-production pattern for energy systems under uncertainty through introducing carbon emission trading and green certificate schemes. *Journal of Cleaner Production*, 161, 299–316, doi: 10.1016/j.jclepro.2017.05.123.

- Tang, L, JR Shi and Q Bao (2016). Designing an emissions trading scheme for China with a dynamic computable general equilibrium model. *Energy Policy*, 97, 507–520, doi: 10.1016/j.enpol.2016.07.039.
- Tanpaifang.com (2016). After 2020, China may impose a carbon tax. Available at <http://www.tanpaifang.com/tanshui/2016/0810/55358.html>. Accessed on December 2017.
- The Central People's Government of the People's Republic of China (2017). National Population development plan (2016–2030). Available at <http://www.nhfpc.gov.cn/bgt/gwywj2/201701/a6dd45cf1660432c91a9ada5e758a4a2.shtml>. Accessed on December 2017.
- Venmans, F (2012). A literature-based multi-criteria evaluation of the EU ETS. *Renewable and Sustainable Energy Reviews*, 16(8), 5493–5510, doi: 10.1016/j.rser.2012.05.036.
- Viskovic, A, V Franki and V Valentic (2014). CCS (carbon capture and storage) investment possibility in South East Europe: A case study for Croatia. *Energy*, 70, 325–337, doi: 10.1016/j.energy.2014.04.007.
- Viskovic, V, YS Chen and AS Siddiqui (2017). Implications of the EU emissions trading system for the South-East Europe regional electricity market. *Energy Economics*, 65, 251–261, doi: 10.1016/j.eneco.2017.04.033.
- Wang, T, C Ma and C Lin (2012). SG-RAS/SG-CE methods for SAM balancing. *Statistical Research*, 29(12), 88–95.
- Wang, ZX, J Zhao and M Li (2017). Analysis and optimization of carbon trading mechanism for renewable energy application in buildings. *Renewable and Sustainable Energy Reviews*, 73, 435–451, doi: 10.1016/j.rser.2017.01.094.
- Wu, R, HC Dai, Y Geng, Y Xie, T Mosui and X Tian (2016). Achieving China's INDC through carbon cap-and-trade: Insights from Shanghai. *Applied Energy*, 184, 1114–1122, doi: 10.1016/j.apenergy.2016.06.011.
- Yang, HX, JW Luo and HJ Wang (2017). The role of revenue sharing and first-mover advantage in emission abatement with carbon tax and consumer environmental awareness. *International Journal of Production Economics*, 193, 691–702, doi: 10.1016/j.ijpe.2017.08.032.
- Zetterberg, L (2014). Benchmarking in the European Union Emissions Trading System: Abatement incentives. *Energy Economics*, 43, 218–224, doi: 10.1016/j.eneco.2014.03.002.
- Zhang, CH, WM Ju, JM Chen, M Zan, DQ Li, YL Zhou and XQ Wang (2013). China's forest biomass carbon sink based on seven inventories from 1973 to 2008. *Climate Change*, 118(3–4), 933–948, doi: 10.1007/s10584-012-0666-3.
- Zhang, K, Q Wang, QM Liang and H Chen (2016). A bibliometric analysis of research on carbon tax from 1989 to 2014. *Renewable and Sustainable Energy Reviews*, 58, 297–310, doi: 10.1016/j.rser.2015.12.089.
- Zhu, L, XB Zhang, Y Li, X Wang and JX Guo (2017). Can an emission trading scheme promote the withdrawal of outdated capacity in energy-intensive sectors? A case study on China's iron and steel industry. *Energy Economics*, 63, 332–347, doi: 10.1016/j.eneco.2017.02.004.